

Ram Reddy

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PROFESSIONAL SUMMARY

Data Analyst with over 4 years of experience and a proven track record as a Subject Matter Expert (SME) in transforming complex data into actionable insights, driving operational efficiencies, and ensuring robust data governance practices across industries. Proficient in Tableau, Power BI, SQL, and Python libraries like Pandas, NumPy, and Matplotlib for data visualization, manipulation, and statistical analysis. Expertise in data preprocessing, cleaning, and integrating large datasets, ensuring accuracy and optimizing data retrieval processes. Skilled in financial forecasting, statistical analysis, and data storytelling, effectively communicating insights to non-technical stakeholders and driving business solutions.

WORK EXPERIENCE

Data Analyst, Merck

Sep 2023 - Present

- Extracted, cleaned, and preprocessed large-scale pharmacovigilance datasets using SQL and PostgreSQL, ensuring accuracy and consistency of over 3 million records from global regulatory sources.
- Developed interactive dashboards in Tableau for real-time monitoring of ADR trends and safety metrics, reducing reporting delays by 15% and enabling data-driven decisions for senior management.
- Developed machine learning models using Python (scikit-learn, pandas, NumPy) to predict adverse drug reactions (ADRs) from historical pharmacovigilance data, achieving an 80% accuracy rate in identifying potential safety concerns early.
- Collaborated with cross-functional teams (clinical research, regulatory affairs, pharmacovigilance) to define project scope and integrate data from clinical trials, post-market surveillance, and spontaneous reporting systems, speeding up decision-making.
- Applied advanced statistical methods such as survival analysis and regression to identify factors contributing to ADRs, improving the identification of high-risk patient subgroups, and achieving a 10% improvement in regulatory compliance.
- Utilized AWS Redshift and S3 for efficient data storage, enabling scalable processing of large pharmacovigilance datasets and reducing data retrieval times by 30%.
- Implemented time-series forecasting models (ARIMA, Exponential Smoothing) for predicting future ADR trends, enabling proactive risk management, and potentially saving \$2 million in post-market regulatory penalties and recalls.
- Developed A/B testing frameworks to evaluate the effectiveness of pharmacovigilance interventions, resulting in a 12% improvement in ADR detection accuracy and faster safety response times.
- Enhanced data integrity and accuracy through rigorous data cleaning and preprocessing using Python (pandas, NumPy), achieving a 99.5% accuracy rate in final ADR datasets, improving the reliability of subsequent safety analyses.

Data Analyst, Capgemini

May 2019 - Aug 2022

- Developed robust financial forecasting models using Python, achieving a variance of less than 5% between projected and actual financial outcomes. Collaborated with IT teams to enhance data warehouse capabilities, ensuring seamless integration.
- Applied advanced statistical techniques to analyze customer behavior, leading to an 8% increase in customer retention rates.
- Implemented AWS data solutions to optimize financial data management, leveraging services such as Amazon S3 for secure data storage and AWS Glue for efficient ETL processes, resulting in a 35% improvement in data retrieval efficiency.
- Engineered and maintained a sophisticated Power BI dashboard for real-time financial monitoring, facilitating strategic decision-making and providing stakeholders with actionable insights.
- Contributed to the development and enhancement of advanced risk assessment models, improving the organization's ability to effectively mitigate financial risks and ensuring regulatory compliance.
- Optimized complex SQL databases by implementing advanced techniques such as window functions and CTEs, ensuring efficient data storage, retrieval, and integration with analytical tools, thereby enhancing overall data accessibility and performance.
- Utilized data storytelling to communicate insights effectively to non-technical stakeholders, driving continuous improvement through Agile methodologies. Led data mapping and transformation initiatives to deliver accurate and timely financial reporting for stakeholders at all organizational levels.
- Automated routine financial reporting tasks in Excel by creating custom macros and VBA scripts, improving the efficiency and accuracy of monthly reports, and reducing manual errors by 25%.

TECHNICAL SKILLS

Methodologies: SDLC, Agile, Waterfall.

Languages: Python, SQL, R, Scala, Spark, Hadoop, SAS.

Packages: Pandas, NumPy, Matplotlib, SciPy, Seaborn, Scikit-Learn, ggplot2, TensorFlow.

Data Visualization Tools: Tableau, Power BI, Looker, AWS Quicksight, Advanced Excel, Statistics.

Databases: MySQL, MongoDB, PostgreSQL, SQL Server.

Cloud Technologies & IDEs: Amazon Web Services, Microsoft Azure, Visual Studio Code, PyCharm, Jupyter Notebook, RStudio.

Data Manipulation: Data Analysis, Data Mining, Data Preprocessing, Data Mapping, Data Cleaning, Data Visualization, Data Modeling, Data Warehousing, Data Storytelling, Data Wrangling, Data Acquisition, Data Integration, Data Transformation.

Other Technical Skills: SSIS, SSRS, Machine Learning, CI/CD, Probability distributions, Confidence Intervals, ANOVA, Clustering, Advanced Analytics, ETL Processes, Informatica, Report Generation, Statistical Analysis, Regression Analysis, A/B Testing, Forecasting & Modeling, Hypothesis Testing, Time Series Analysis, RapidMiner, OLAP, KPI tracking, metadata management, data lineage, master data management (MDM), neural networks, Natural Language Processing (NLP), custom macros and VBA scripts.

EDUCATION

University of North Texas

Master of Science, Computer Science

Sep 2022 - May 2024

GPA: 3.65/4.0